

Seismic Performance with Fluid Viscous Dampers

Individual academic project | University of Windsor | 2025

A three-storey reinforced-concrete frame was modelled in ETABS to compare damper layouts and total damping coefficients under two earthquake records.

3	storeys
2	ground motions
600-1,800	kN-s/m tested
30-40%	typical Northridge reduction

QUESTION

How do linear viscous-damper placement and total damping coefficient affect seismic response in a small reinforced-concrete frame?

METHOD

- Linear direct-integration time-history analysis in ETABS.
- Loma Prieta 1989 and Northridge 1994 ground motions.
- Baseline, single-base-damper, and three-distributed-damper configurations.
- Total C values of 600, 1,200, and 1,800 kN-s/m.

EVIDENCE

Typical response reductions were approximately 18-20% for Loma Prieta and 30-40% for Northridge, with some Northridge responses approaching 50% reduction.

ENGINEERING DECISION

The distributed layout with a total coefficient of 1,800 kN-s/m produced the strongest overall response control among the modelled cases.

LIMITATIONS

Results are specific to the linear analytical model, selected records, scaling assumptions, and tested layouts; they are not a design recommendation for a real structure.

Evidence basis: course model, result tables, MATLAB processing, presentation, and public project repository. This is a portfolio summary, not a sealed engineering document.